

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/381251423>

Transcriptomic responses of XDR *Klebsiella pneumoniae* to N-acetyl cysteine reveals suppression of major biogenesis pathways leading to bacterial killing and biofilm eradication

Article in *Journal of Applied Microbiology* · June 2024

DOI: 10.1093/jambio/txae136

CITATION

1

READS

143

4 authors, including:



Ankurita Bhowmik
Patanjali Research Foundation

10 PUBLICATIONS 33 CITATIONS

[SEE PROFILE](#)



Sambuddha Chakraborty
Tripura University

13 PUBLICATIONS 65 CITATIONS

[SEE PROFILE](#)



Ashwini Chauhan
University of Delhi

62 PUBLICATIONS 2,105 CITATIONS

[SEE PROFILE](#)

Transcriptomic responses of extensively drug resistant *Klebsiella pneumoniae* to N-acetyl cysteine reveals suppression of major biogenesis pathways leading to bacterial killing and biofilm eradication

Ankurita Bhowmik^{1,‡}, Sambuddha Chakraborty^{1,3,‡}, Anusha Rohit², Ashwini Chauhan^{1,3,*}

¹Department of Microbiology, Tripura University, Suryamaninagar, Agartala, West Tripura 799022, India

²Department of Microbiology, Madras Medical Mission Hospital, Chennai, Tamil Nadu 600037, India

³Department of Microbiology, BK Bachhawat Block, University of Delhi South Campus, Benito Juarez Road, New Delhi 110021, India

*Corresponding author. Department of Microbiology, BK Bachhawat Block, University of Delhi South Campus, Benito Juarez Road, New Delhi 110021, India.

E-mail: ashwinichauhan@south.du.ac.in

[‡]Equal contributions.

Abstract

Aims: Carbapenemase-producing *Klebsiella pneumoniae* is categorized as a “critical global priority-one” pathogen by WHO and new and efficient treatment options are warranted. This study aims to assess the antibacterial and antibiofilm potential of N-acetyl cysteine (NAC), against clinical isolates of extensively drug resistant (XDR) *K. pneumoniae* and elucidate the mechanism of killing.

Methods and results: XDR-*K. pneumoniae* were isolated from patients admitted to Madras Medical Mission Hospital, India. Antibiofilm activity of NAC was checked using *in vitro* continuous flow model and RNA sequencing was done using Illumina Novoseq. Data quality was checked using FastQC and MultiQC software. Our findings revealed that NAC at a concentration of 100 mg/ml was safe, and could inhibit the growth and completely eradicate mature biofilms of all XDR-*K. pneumoniae* isolates. Transcriptomic responses in XDR-*K. pneumoniae* to NAC showed significant downregulation of the genes associated with crucial biogenesis pathways, including electron transport chain and oxidoreductase activity besides a specific cluster of genes linked to ribosomal proteins.

Conclusions: Our results indicate that NAC kills the XDR-*K. pneumoniae* clinical isolates by shutting the overall metabolism and, hence, successfully eradicate *in vitro* biofilms formed on catheters.

Impact Statement

N-acetyl cysteine provides a safe and efficient alternative to antibiotics against extensively drug resistant-*Klebsiella pneumoniae* infections

Keywords: *Klebsiella pneumoniae*; multidrug resistance; Biofilm; N-acetyl cysteine; RNA Sequencing

Introduction

Klebsiella pneumoniae is a major cause of community- and hospital-acquired infections in immunocompetent and immunocompromised patients (Joseph et al. 2021). Moreover, the propensity of *K. pneumoniae* to acquire multidrug resistance has made them more notorious to deal with in the hospital environment. WHO-published global priority pathogen list in 2017 has included ESBL-producing carbapenem-resistant *K. pneumoniae* among the “critical priority pathogens” (WHO 2017). In India, *K. pneumoniae* was recently reported to be prevalent in up to 18% of the clinical samples analyzed. Furthermore, the Indian Council of Medical Research (ICMR) has also disclosed an extended resistance in *K. pneumoniae* to cephalosporins, aminoglycosides, carbapenems, fluoroquinolones, and polymyxin (Indian Council of Medical Research 2020) and thus has classified it as a Group I pathogen. In addition to different resistance mechanisms, the biofilm formation ability of *K. pneumoniae* confers additional protection against antibiotics and the host immune system (Guilhen et al. 2019).

Globally, the rise and spread of the extensively drug resistant (XDR) *K. pneumoniae* infections have renewed the research interest in studying the correlation between antibiotic resistance and biofilm-encoding genes in *K. pneumoniae* (Efah et al. 2020). Currently, different combinations of antibiotics, such as meropenem with fosfomycin and/or gentamicin, ceftazidime/avibactam with fosfomycin, gentamicin, or tigecycline, etc. are used to treat infections of XDR-*K. pneumoniae* (Daikos et al. 2014). In addition, alternatives to antibiotics, such as bacteriophage therapy in conjunction with nonactive antibiotics, such as sulfamethoxazole-trimethoprim, have also been explored, which were effective against the isolates of XDR-*K. pneumoniae* (Bao et al. 2020). Recent reports demonstrated the potential of antimicrobial peptides such as AA139, SET-M33, and ΔM2 against colistin-resistant *K. pneumoniae* (van der Weide et al. 2019). EDTA in combination with colistin has recently been reported to treat colistin-resistant *K. pneumoniae* (Shein et al. 2021). However, to manage the drug resistance challenge among *K. pneumoniae*, constant exploration of new non-antibiotic therapeutic interventions is urgently required.

Received 3 November 2023; revised 12 April 2024; accepted 5 June 2024

© The Author(s) 2024. Published by Oxford University Press on behalf of Applied Microbiology International. All rights reserved. For permissions, please e-mail: journals.permissions@oup.com

N-acetyl cysteine (NAC) is a synthetic derivative of endogenous amino acid, L-cysteine. It is an FDA-approved mucolytic drug used for decades to treat acetaminophen overdose and obstructive pulmonary diseases (Tenorio et al. 2021). Due to its mucolytic activity, NAC has attracted microbiologists to explore its potential as an antibacterial and antibiofilm agent. Several *in vitro* studies evaluated the activity of NAC against different Gram-positive and Gram-negative bacteria, including *Escherichia coli*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, *K. pneumoniae*, *Staphylococcus aureus*, and *Staphylococcus epidermidis*, among others (Jun et al. 2019). Given the broad-spectrum activity of NAC against bacteria, this study aims to repurpose NAC to treat biofilms of XDR-*K. pneumoniae*, isolated from patients in hospitals. To date, very few studies have explored the mode of action of NAC. Transcriptome analysis offers the advantage of comprehensively understanding genetic modulation within the cell and highlights crucially impacted gene networks. Moreover, the increasing challenge of antibiotic resistance has left us with a diminishing pool of novel target sites for designing new antibiotics. Transcriptome analysis provides pivotal insights, allowing for a transcriptomics-based drug repositioning approach and the identification of new target proteins (Lim et al. 2021). Therefore, we have chosen to conduct transcriptome analysis to gain insight into the genetic portfolio of NAC-treated bacteria. In our study, based on the prevalence and resistance profile, we selected 12 XDR-*K. pneumoniae* isolates and assessed the *in vitro* antibacterial as well as antibiofilm activity of NAC against these isolates. We observed that NAC was bactericidal against all the XDR-*K. pneumoniae* isolates at a concentration of 10 mg/ml. Furthermore, NAC could successfully eradicate *in vitro* biofilms, grown on TIVAP, of all XDR-*K. pneumoniae* at an acceptable concentration of 100 mg/ml (NIH U.S. National Library of Medicine 2018). Next, we used RNA-seq methods to investigate how NAC inhibits XDR-*K. pneumoniae*. The transcriptome data demonstrated that NAC administration dramatically reduced a range of enzymes involved in several biogenesis processes, such as *K. pneumoniae* oxidoreductase activity, menaquinone biosynthesis, histidine biosynthesis, uridine nucleosidase, and pyridoxine biosynthesis. As safe antibiotic alternatives are urgently needed to treat *K. pneumoniae* infections and enhance treatment outcomes. These findings contribute to a better knowledge of NAC's antibacterial actions against *K. pneumoniae* and its safe usage in the treatment of XDR-*K. pneumoniae*, will lessen the socio-medico-economic burden.

Materials and methods

Bacterial strains and growth media

All XDR-*K. pneumoniae* isolates were grown at 37°C in Luria-Bertani broth (LB) purchased from Himedia, India (M1245). For viable cell estimation, the serially diluted culture was plated on sterile LB agar plates and incubated overnight at 37°C.

Antimicrobial agents

NAC was purchased from Sigma-Aldrich (A7250). The isotonic NAC solution was prepared by dissolving NAC in 0.9% sterile saline.

Estimation of antibiotic susceptibility

VITEK 2 compact (bioMérieux, France) was used for biochemical characterization and determining the antibiogram of all the XDR-*K. pneumoniae* isolates using disposable antibiotic susceptibility testing cards. Wells with no antibiotics served as the positive control.

Assessing the minimum inhibitory concentration

The minimum inhibitory concentration (MIC) of NAC against the clinical isolates of XDR-*K. pneumoniae* was determined by broth microdilution assay (Lebeaux et al. 2014). Briefly, NAC was serially diluted in LB in a microtiter plate to give a 2-fold serial dilution. An equal volume of exponentially growing *K. pneumoniae* isolate was added to give a final inoculum of 5×10^5 CFU/ml in all wells except for the negative control. Well with bacteria but no NAC, served as an untreated control. The plate was incubated for 18 h at 37°C under static conditions and the first well without visible growth was noted as the MIC concentration. All experiments were carried out in at least in triplicate.

Effect of NAC on exponential phase planktonic XDR-*K. pneumoniae*

The efficacy of NAC in inhibiting growth against clinical isolates of XDR-*K. pneumoniae* was evaluated on microtiter plates using a Biotek Epoch-2 microtiter plate reader (Agilent Technologies, USA). Each treatment was replicated in at least 3 wells. Log phase culture ($OD_{600} \approx 0.3-0.5$) of *K. pneumoniae* isolates was treated with sub-MIC (2.5 or 5 mg/ml) concentration at 80 mg/ml of NAC for 24 h at 37°C with continuous shaking. The untreated wells served as a control. The OD_{600nm} were measured by the Biotek Epoch-2 plate reader for 24 h. After 24 h, the viable cell count was estimated by plating the serial dilutions on LB agar media, and plates were incubated at 37°C overnight (Lebeaux et al. 2014).

In vitro antibiofilm activity of NAC against the biofilms of XDR-*K. pneumoniae* isolates using continuous flow set up

In vitro antibiofilm assays were first done in microtiter plate as described by Lebeaux et al. (2014) with slight modifications. Overnight cultures of *K. pneumoniae* isolates were inoculated in fresh medium at a 1:100 dilution in a 96-well microtiter plate and incubated for 96 h to form mature biofilm at 37°C without shaking. Planktonic bacteria were removed by gentle pipetting and the wells were rinsed three times with 0.9% saline. The mature biofilms that developed for 96 h in microtiter plates were treated with 125 µl isotonic solution of different concentrations of NAC (5, 10, 20, 40, 80, and 100 mg/ml). The treatment was allowed to dwell in the wells for 24 h at 37°C under static conditions. 0.9% saline-treated wells served as controls. After 24 h, the NAC solution was removed from the wells and rinsed thrice with 0.9% saline. Wells were scraped vigorously to remove any residual biofilm and homogenized with 0.9% saline. The viable cell count was estimated by plating the serial dilutions on LB agar medium and the plates were incubated at 37°C. Each treatment was given in replicates of four and each experiment was repeated at least thrice.

Further, *in vitro* efficacy of NAC against the device-associated biofilms of 4 XDR-*K. pneumoniae* isolates was assessed by using a previously described continuous flow system (Chauhan et al. 2012). Optimum bacterial cells (10^2 cells/100 μ l) were injected through the silicon septum of a totally implantable venous access ports (TIVAP) catheter, and the bacterial cells were allowed to adhere to the TIVAP surface for 3 h at 37°C. Once adhered, the continuous flow system was turned on at a speed of 300 μ l/min for 24 h to allow the growth of biofilms inside the TIVAP. The non-adherent cells and the used media were collected in the waste container. After that, the formed biofilms were treated with 100 mg/ml NAC for 24 h under static conditions by instilling the NAC into the TIVAP lumen, and the cell viability was checked by CFU/ml plating. Untreated TIVAP-associated biofilms were used as control.

RNA-seq analysis

The RNA of NAC-treated *K. pneumoniae* samples was extracted using HiMedia HiPurA bacterial RNA extraction kit and eluted with 16 μ l of RNase-free water. The RNA quality assessment was done using RNA Screen Tape System (Catalog: 5067-5576, Agilent) in a 4150 Tape Station System (Catalog: G2992AA, Agilent). RNA concentration was determined on Qubit® 3.0 Fluorometer (Catalog: Q33216, Thermo Fisher Scientific) using the Qubit™ RNA BR Assay Kit (Catalog: Q32853, Thermo Fisher Scientific). The sequence data was generated using Illumina Novoseq. Data quality was checked using FastQC (Andrews 2017) and MultiQC (Ewels et al. 2016) software. The data was checked for base call quality distribution, % bases above Q20, Q30, %GC, and sequencing adapter contamination. All the samples have passed the QC threshold (Q20 > 95%). Raw sequence reads were processed to remove adapter sequences and low-quality bases using fastp. The QC-passed reads were mapped onto indexed *K. pneumoniae* subsp. *pneumoniae* HS11286 reference genome using Bowtie v2 (Langmead and Salzberg 2012) aligner. Genes with absolute log2 fold change ≥ 1 and *P*-value ≤ 0.05 were considered significant. The expression profile of differentially expressed genes across the samples is presented in volcano plots. The genes that showed significant differential expression were used for gene ontology and pathway enrichment analysis. Enrichment analysis for biological processes, molecular functions, cellular components, and the KEGG pathway was performed using DAVID (Huang da et al. 2009).

Statistical analysis

All the results are replicas of at least three independent experiments. Statistical differences were evaluated using one-way analysis of variance (ANOVA) using GraphPad Prism version 10.0. *P* values <0.05, were considered statistically significant.

Results

Klebsiella pneumoniae isolated from clinical settings are resistant to different classes of antibiotics

The characterization and antibiotic susceptibility profiling of the *K. pneumoniae* isolates were performed using an automated Vitek 2 compact system. All *K. pneumoniae* isolates

showed resistance to multiple classes of antibiotics (Fig. 1a). It is concerning to note that all isolates showed 100% resistance to ciprofloxacin, ofloxacin, tigecycline, amikacin, amoxicillin-clavulanate, cefazolin, gentamicin, ceftriaxone, cefuroxime, and cotrimoxazole. Furthermore, <10% of isolates were susceptible to imipenem, meropenem, ertapenem, pip-tazobactam, and cefoperazone sulbactam. Most of the isolates (90%) showed resistance toward tigecycline, netilmicin, and ceftipime.

Colistin is currently used as a last resort to treat multidrug-resistant *K. pneumoniae*. We also observed that 27% of *K. pneumoniae* isolates were resistant to colistin, which is concerning and indicative of the probable spread of colistin resistance genes. A recent report in 2021 showed the tendency of carbapenem-resistant *K. pneumoniae* to acquire resistance to colistin under tigecycline or colistin pressure (Lebeaux et al. 2014). A similar trend was observed in our study, where all the colistin resistant *K. pneumoniae* isolates were also resistant to carbapenems (Fig. 1a).

NAC effectively kills clinical isolates of XDR-*K. pneumoniae*

To assess the antibacterial efficacy of NAC, we determined its MIC against clinical isolates of XDR-*K. pneumoniae* according to the CLSI guidelines. Interestingly, we observed ~83% (10 out of 12) of the isolates of XDR-*K. pneumoniae* that we chose for the study were susceptible to 5 mg/ml of NAC. The remaining 2 isolates of XDR-*K. pneumoniae* were susceptible to a two-fold higher concentration of NAC, that is, 10 mg/ml (Supplementary Table S1). Furthermore, we checked the effect of NAC on the cell viability of the isolates of XDR-*K. pneumoniae* by treating exponentially grown cells with NAC from sub-MIC concentration (2.5 or 5 mg/ml) to 80 mg/ml for 24 h and checked changes in growth kinetics and colony-forming units (CFU) (Table 1) (Supplementary Fig. S1). We observed that NAC was bactericidal at MIC concentration against all the clinical isolates except KPO760. At MIC (5 mg/ml), NAC reduced the growth of KPO760 up to 4.8-logs, which completely disappeared with a two-fold increase in MIC (10 mg/ml). The minimum limit of deduction in Table 1 is 1 CFU/ml. The absence of any colonies upon plating the entire volume confirmed the eradication of viable cells following treatment.

NAC effectively eliminates TIVAP-associated mature biofilms of XDR-*K. pneumoniae*

Managing biofilm-associated infections is challenging, especially when infections are due to antibiotic-resistant isolates such as XDR-*K. pneumoniae*. We evaluated the *in vitro* antibiofilm activity of NAC against 12 XDR-*K. pneumoniae* isolates. The 96 h preformed biofilms were treated with different concentrations of NAC prepared in an isotonic solution, ranging from 5 to 100 mg/ml, for 24 h, and the cell viability was estimated by CFU/ml plating (Table 2).

The mature biofilms of KPON2697 and KPO3875 were completely eradicated by using 10 mg/ml NAC solution and biofilms of the XDR isolate KPO3337 were eradicated by 20 mg/ml NAC solution (Supplementary Fig. S2). A concentration-dependent antibiofilm activity of NAC was observed against other XDR-*K. pneumoniae* isolates. Although 80 mg/ml NAC solution reduced the biofilm cell viability to 6-log (99.999%), some viable cell populations were still

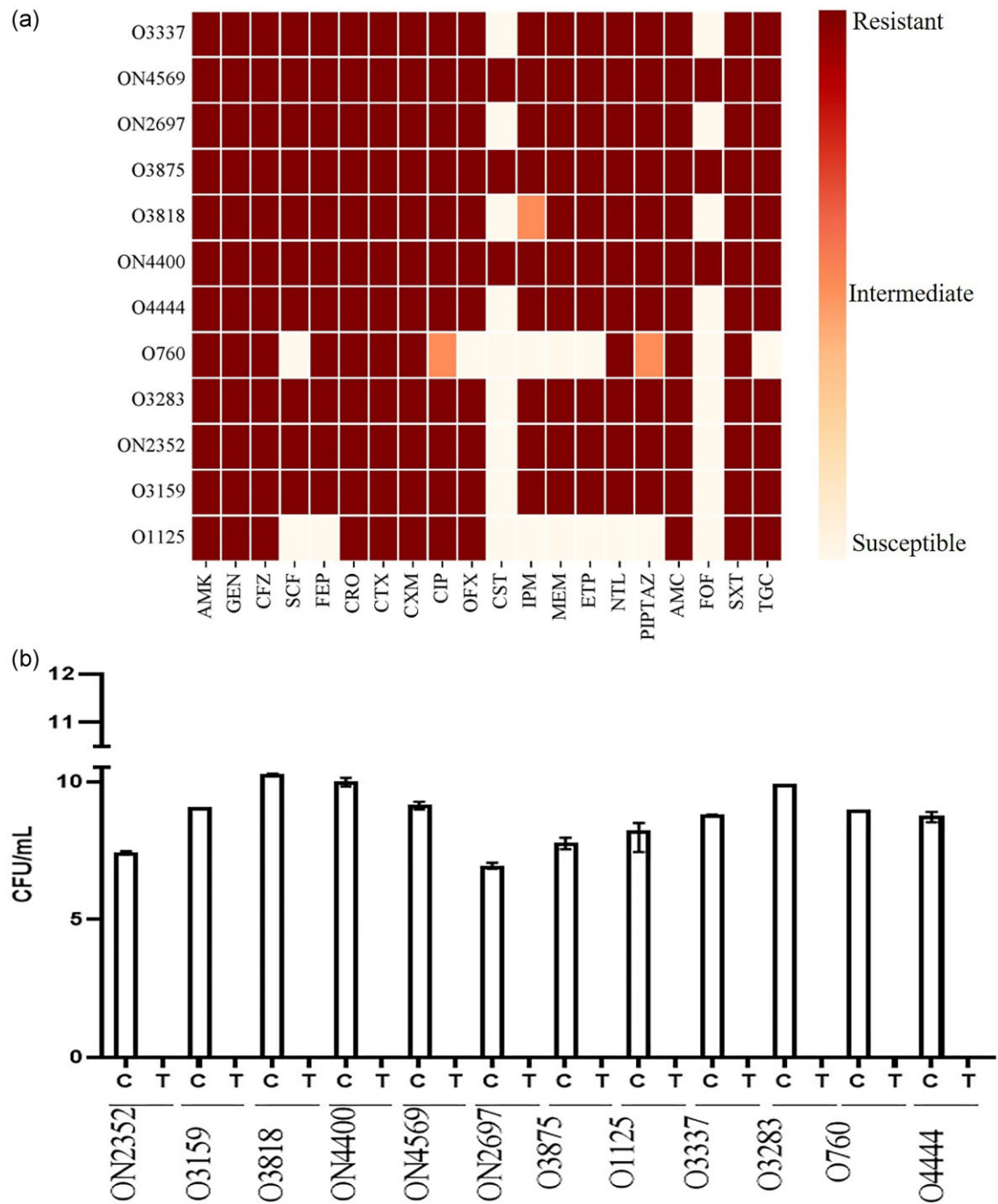


Figure 1. NAC could successfully eradicate multidrug-resistant *K. pneumoniae* clinical isolates. (a) Drug resistance profile of 12 XDR-*K. pneumoniae* isolates used in this study. The heatmap represents the antibiogram and extensive antibiotic resistance among the *K. pneumoniae* isolates. Rows represent *K. pneumoniae* isolates and columns represent different antibiotics. AMK-Amikacin, GEN-Gentamicin, CFZ-Cefazolin, SCF-Cefoperazone sulbactam, FEP-Cefepime, CRO-Ceftriaxone, CTX-Cefotaxime, CXM-Cefuroxime, CIP-Ciprofloxacin, OFX-Ofloxacin, CST-Colistin, IPM-Imipenem, MEM-Meropenem, ETP-Ertapenem, NTL-Netilmycin, PIP-TAZ-Pip tazobactam, AMC-Amoxicillin clavulanate, FOF-Fosfomycin, SXT-Co-trimoxazole, TGC-Tigecycline. (b) *In vitro* biofilms of *K. pneumoniae* isolates were grown in TIVAP using a continuous flow system and treated with NAC (100 mg/ml) for 24 h. Viable cell count was analyzed by plating CFU/ml ($n = 4$ for each isolate) on LB agar plates. (P value < 0.001). Abbreviations: C, control, and T, treated. Scale of Y axis is in log10.

recovered. Interestingly, 100 mg/ml NAC solution totally eradicated the biofilms formed by all the isolates, after 24 h treatment (Table 2). Furthermore, biofilms were allowed to form inside the TIVAP for 24 h and were treated with 100 mg/ml NAC for 24 h. In contrast to the untreated control, NAC at 100 mg/ml

completely eliminated all biofilms that had developed inside the TIVAP catheter, as indicated by the absence of viable cells on the plates (Fig. 1b). Previous studies have demonstrated that viable cells in a biofilm could reach up to 10 to 11-log CFU while grown using a continuous flow system (Chauhan *et al.* 2012, Lebeaux *et al.* 2014).

Table 1. Effect of different concentrations of NAC on exponentially growing clinical isolates of XDR-*K. pneumoniae*.

NAC treatment (mg/ml)							
<i>Klebsiella pneumoniae</i> clinical isolates	Untreated	2.5	5	10	20	40	80
(Log reduction)							
ON2352	8.04 (SD: 7.47)	3.88 (SD: 3.37)	0 (MIC)	0	0	0	0
O3159	9.43 (SD: 8.72)	ND*	3.72 (SD: 3.38)	0 (MIC)	0	0	0
O3818	9.93 (SD: 9.38)	6.00 (SD: 5.37)	0 (MIC)	0	0	0	0
ON4400	7.08 (SD: 6.36)	6.67 (SD: 5.79)	0 (MIC)	0	0	0	0
ON4569	9.98 (SD: 9.38)	6.11 (SD: 5.52)	0 (MIC)	0	0	0	0
O4444	10.15 (SD: 9.25)	5.66 (SD: 5.32)	0 (MIC)	0	0	0	0
ON2697	10.40 (SD: 9.66)	7.08 (SD: 6.62)	0 (MIC)	0	0	0	0
O3337	9.93 (SD: 9.38)	5.98 (SD: 5.18)	0 (MIC)	0	0	0	0
O3875	8.04 (SD: 7.37)	5.00 (SD: 4.62)	0 (MIC)	0	0	0	0
O1125	9.75 (SD: 8.78)	ND*	4.81 (SD: 3.70)	0 (MIC)	0	0	0
O3283	7.98 (SD: 6.83)	5.52 (SD: 4.59)	0 (MIC)	0	0	0	0
O760	9.41 (SD: 6.29)	7.15 (SD: 6.29)	5.08 (MIC; SD: 4.50)	0	0	0	0

*ND: Not done.

Concentration of NAC is 2.5mg/mL to 100mg/mL. *K. pneumoniae* clinical isolates are designated as ON or O followed by numbers.**Table 2.** Antibiofilm activity NAC against viable biofilm cells of XDR-*K. pneumoniae* isolates.

NAC concentration(mg/ml)							
<i>Klebsiella pneumoniae</i> isolates	Untreated	5	10	20	40	80	100
(Log reduction)							
ON2352	8.43 (SD: 6.51)	5.28 (SD: 5.66)	5.23 (SD: 4.91)	3.67 (SD: 4.38)	3.52 (SD: 3.55)	3.20 (SD: 2.91)	0.00
O3159	9.08 (SD: 8.63)	7.04 (SD: 6.47)	6.56 (SD: 6.38)	6.15 (SD: 6.15)	4.80 (SD: 5.87)	3.66 (SD: 5.54)	0.00
O3818	10.28 (SD: 8.64)	9.11 (SD: 8.44)	8.00 (SD: 8.24)	7.98 (SD: 7.32)	6.63 (SD: 6.47)	4.75 (SD: 4.42)	0.00
ON4400	10.00 (SD: 9.54)	7.11 (SD: 6.48)	7.00 (SD: 6.40)	6.95 (SD: 6.47)	6.70 (SD: 6.32)	7.63 (SD: 5.37)	0.00
ON4569	9.15 (SD: 8.64)	8.88 (SD: 8.38)	8.66 (SD: 8.38)	7.63 (SD: 7.38)	5.66 (SD: 5.95)	4.04 (SD: 3.18)	0.00
ON2697	6.95 (SD: 6.32)	4.85 (SD: 4.48)	0.00	0.00	0.00	0.00	0.00
O3875	7.80 (SD: 7.45)	5.52 (SD: 6.29)	0.00	0.00	0.00	0.00	0.00
O3337	8.23 (SD: 8.38)	6.00 (SD: 6.32)	5.75 (SD: 4.47)	0.00	0.00	0.00	0.00
O1125	8.80 (SD: 5.39)	7.23 (SD: 5.20)	6.92 (SD: 4.31)	5.28 (SD: 6.47)	4.93 (SD: 5.51)	3.86 (SD: 4.33)	0.00
O760	9.90 (SD: 5.41)	8.54 (SD: 5.48)	6.28 (SD: 7.17)	5.85 (SD: 0.66)	4.18 (SD: 4.40)	3.60 (SD: 5.03)	0.00
O3283	8.98 (SD: 4.48)	8.75 (SD: 5.54)	6.66 (SD: 6.47)	5.11 (SD: 0.37)	4.92 (SD: 8.38)	3.66 (SD: 3.18)	0.00
O4444	8.75 (SD: 8.3)	7.40 (SD: 7.91)	6.18 (SD: 7.69)	5.62 (SD: 0.67)	4.61 (SD: 5.54)	3.75 (SD: 4.18)	0.00

Concentration of NAC is 5 mg/mL to 100 mg/mL. *K. pneumoniae* clinical isolates are designated as ON or O followed by numbers.

NAC is not toxic to L929 and J774A.1 cell lines

NAC is an FDA-approved compound and is considered one of the important ingredients to prepare powerful antioxidant-rich food. We evaluated the cell toxicity by treating the mouse fibroblast cells (L929) and macrophage cells (J774A.1) with 80 or 100 mg/ml NAC for 30 mins. NAC did not exert any toxic effect on the cell lines at both the evaluated concentrations, as indicated by the high rate of cell survival of both the cell types used in the study (Supplementary Fig. S3). J774A.1 showed a 100% survival rate at both tested concentrations. On the same hand, L929 also had a high survival rate of up to 91% survival at both tested concentrations, which may be due to the reasons that NAC enhances the activity of macrophages (Vecchiarelli et al. 1994).

NAC treatment modulates arrays of metabolic pathways and suppresses genes encoding ribosomal proteins

Upon NAC treatment, 627 genes of the XDR-*K. pneumoniae* were significantly ($P \leq 0.05$) differently expressed (log2 fold change ≥ 1); 125 of the genes were upregulated, and the rest were downregulated. The distribution of differentially expressed genes (DEG) is represented through a volcano

plot (Fig. 2a) and the first 20 most significantly upregulated and downregulated genes were further analyzed. Whereas downregulated genes were mainly involved in several biosynthetic enzymes associated with different crucial biochemical pathways such as menaquinone and histidine biosynthesis, upregulated genes were largely associated with the cellular components of outer membrane-bounded periplasmic space, like ion channels, the ABC transport system, and several transfer RNAs (Fig. 2b and Supplementary Tables S2 and S3). The enrichment data delineate a strong downregulation in whole gene products, such as cellular components of the bacterial cytoplasm. NAC treatment negatively altered protein synthesis by down-regulating metabolic pathways and protein translation (Supplementary Fig. S4A–C). Moreover, the functional enrichment analysis strongly indicated that the ribosomal genes are the most significantly altered gene cluster among 627 DEG with an enrichment score of 7.69 (Supplementary Fig. S4A–C and Supplementary Table S4). Bubble plot of the DEG clusters with an enrichment score >1.5 also indicates a highly significant downregulation of ribosomal genes (Fig. 2c). Analysis of ribosome-associated genes across all of the sub-ontology parameters (cellular component, biological process, and molecular function) indicated that a cluster of 30 genes associated with ribosomal

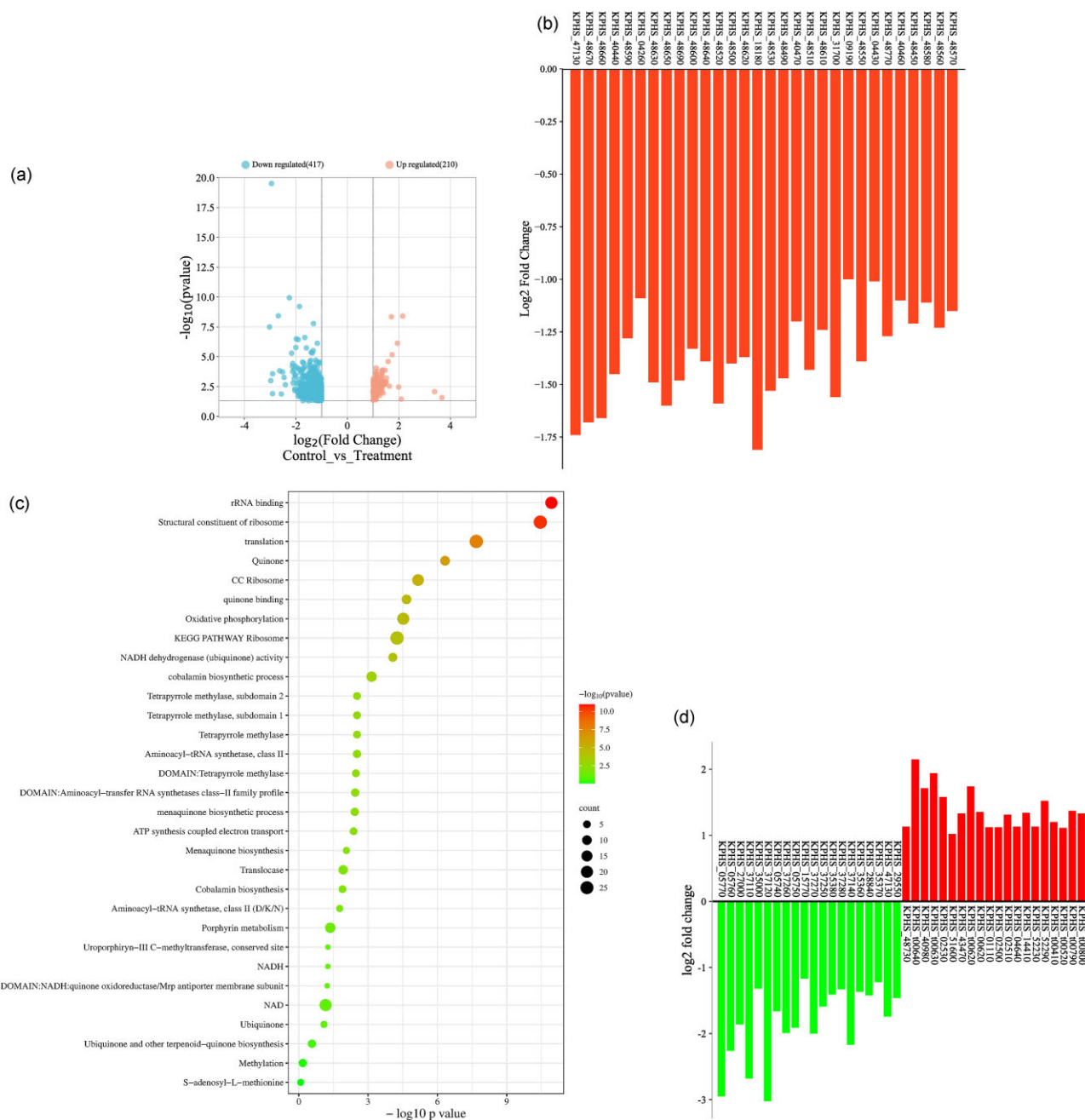


Figure 2. Transcriptomic analysis of XDR *K. pneumoniae* post-NAC treatment. (a) Volcano plot analysis of the 627 DEGs. Teal-colored dots indicate downregulated genes and orange-colored dots indicate upregulated genes ($P \leq 0.05$). (b) Bar plot analysis of 20 most significantly upregulated and downregulated genes. (c) Enrichment bubble plot of gene ontology of DEGs clusters with an enrichment score > 1.5. (d) Bar plot analysis of downregulated 30 ribosomal protein genes.

function were significantly downregulated due to NAC treatment (Fig. 2d and Supplementary Table S5). In all the three groups of gene ontologies, 86.7% of the similar ribosomal protein genes were suppressed, indicating downregulation of discrete ribosomal protein gene cluster due to NAC treatment (Fig. 3a).

Ribosome assembly and function is severely affected in XDR *K. pneumoniae* treated with NAC

We used the STRING database to understand the direct (physical) or indirect (functional) interaction of the proteins encoded by top five significantly downregulated genes involved

in ribosomal function in XDR *K. pneumoniae* treated with NAC, with other proteins.

All these genes encode ribosomal structure proteins (*rplD*: 50S ribosomal protein L4, *rplW*: 50S ribosomal protein L23, *rplS*: 50S ribosomal protein L19, and *rpsQ*: 30S ribosomal protein S17) except *rbfA*, which encodes ribosome-binding factor A. While RbfA revealed a network of five closely associated proteins with a high confidence score, other proteins showed a robust network of ten closely connected proteins. Although RbfA a crucial protein involved in maturation of the 30S ribosomal subunit, it is essential for protein synthesis initiation by binding to mRNA and tRNA.

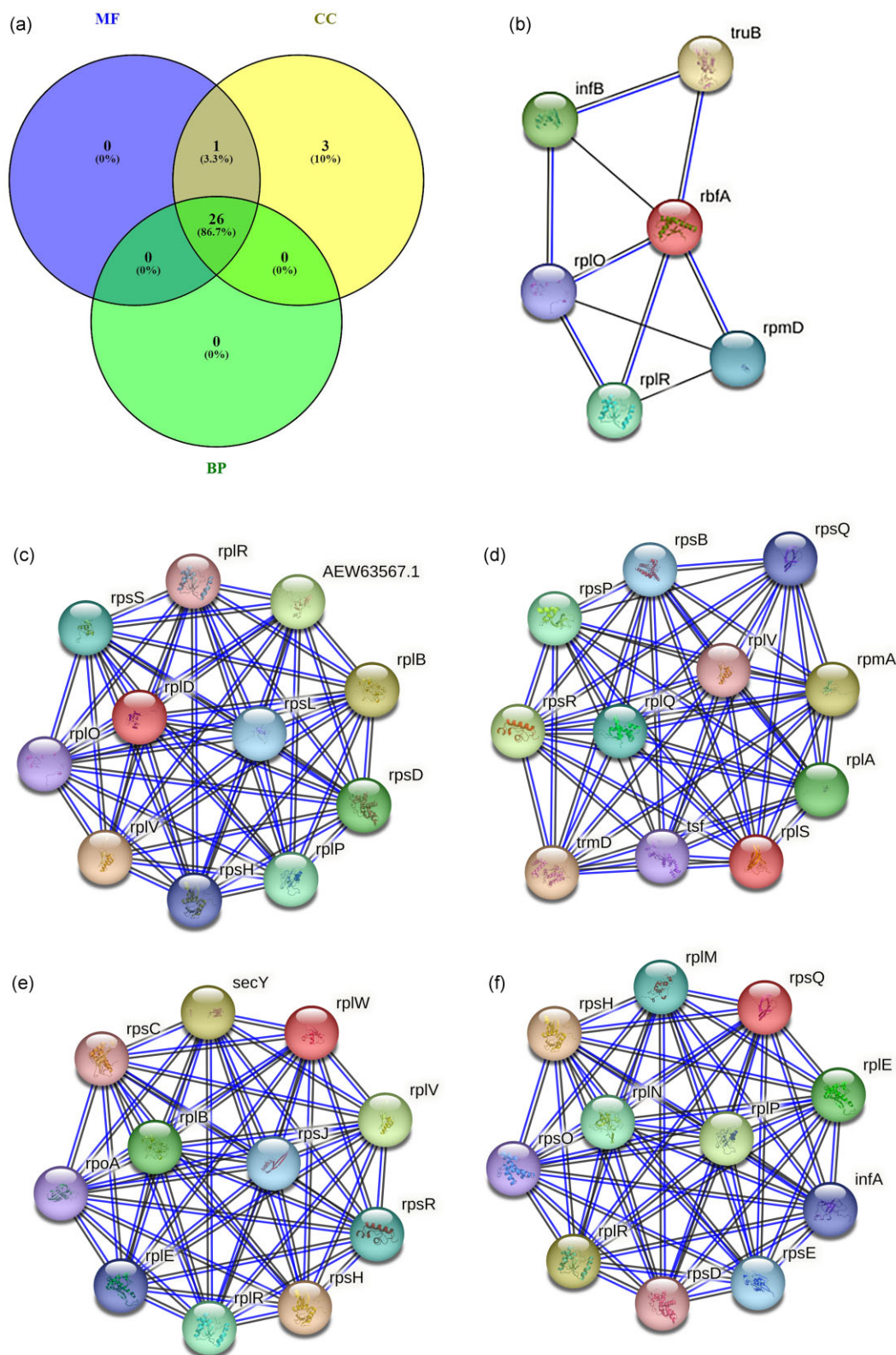
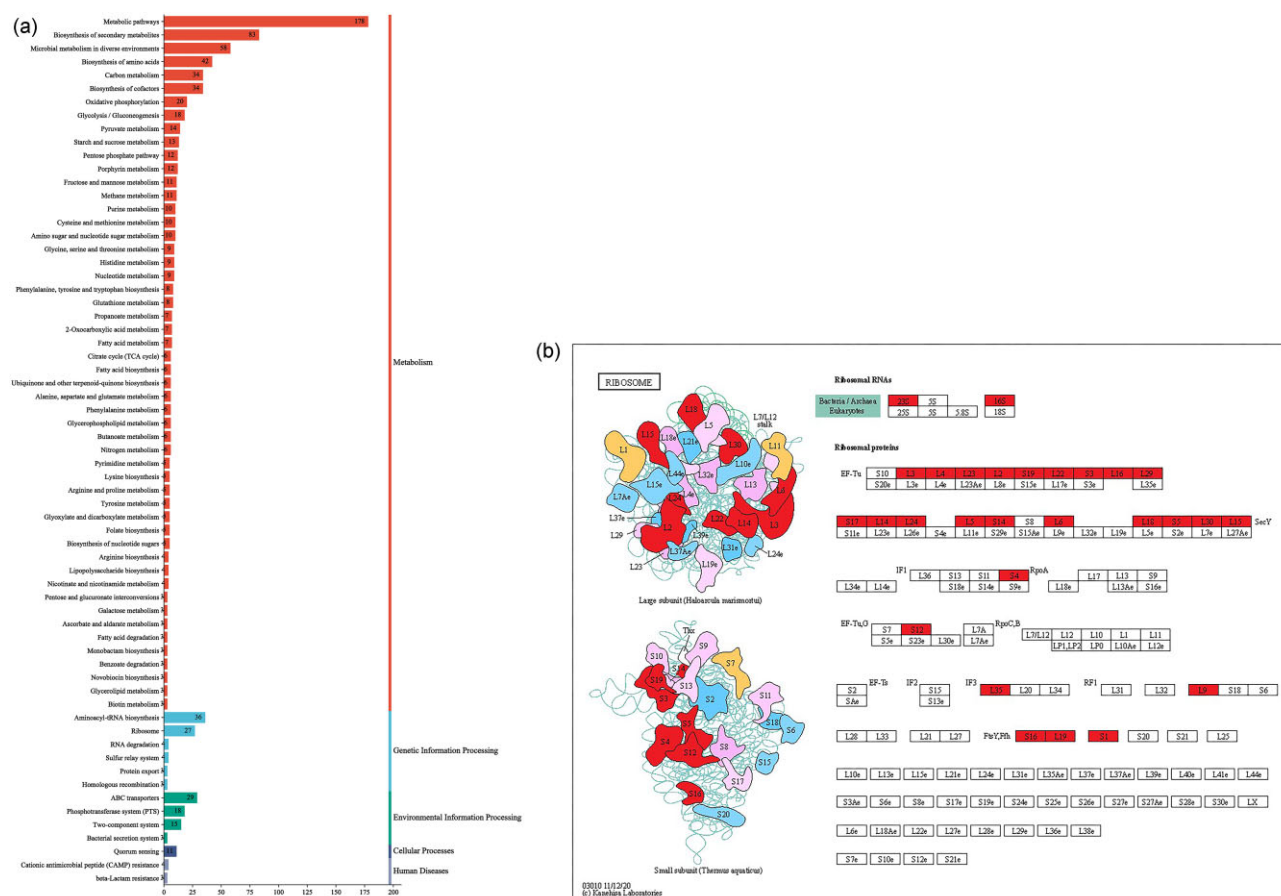


Figure 3. Impact of NAC treatment on ribosomal protein genes. (a) Van diagram of ribosomal genes across the three sub-ontology parameters. (b–f) STRING analysis of five most significantly downregulated ribosomal protein genes (setting: score; high confidence >0.700). Maximum 10 interactions. Active interaction sources: co-expression and co-occurrence. Red-colored node indicates query proteins (KPHS_47130/rbfA, KPHS_48670/rplD, KPHS_48660/rplW, rplS/KPHS_40440, KPHS_48590/rpsQ), and rest of the colored nodes indicate protein networks surfaced in first shells of interaction. For the first query protein among searched ten interactions, we could find five strongly associated protein network with high confidence score, for rest of the query proteins, we found ten most strongly interacted proteins. From the analysis, we can deduce that, except for four proteins (truB, infB, AEW63567.1, and infA), the rest of the associated proteins in the network are either ribosomal large subunit proteins (L1, L2, L6, L10, L11, L7/L12, L13, L15, L16, L17, L18, L22, L23, L24, L27, L29, and L30) or small subunit proteins (S2, S3, S4, S5, S6, S7, S8, S13, S14, S11, S15, S16, S17, and S18) (Supplementary Table S6). Black lines indicate the gene co-expression and the blue line indicates gene co-occurrence.



Numerous studies have highlighted the effectiveness of weak acids in combating bacteria through diverse mechanisms. Notably, research reveals that acetic acid and citric acid, in their neutral forms, possess lipid permeability, facilitating their unrestricted diffusion into microbial cell cytoplasm. This diffusion process achieves equilibrium when the concentrations of the neutral acid inside and outside cells are balanced. Many neutralophilic microorganisms maintain a pH gradient across their cytoplasmic membranes, with the interior typically more alkaline than the exterior. Consequently, this process results in the accumulation of elevated levels of charged weak acid within the cytoplasm, disrupting cellular homeostasis (Hirschfield et al. 2003). NAC is an FDA-approved drug used to treat acetaminophen poisoning and obstructive pulmonary diseases.

In this study, we assessed the potential of NAC against the clinical isolates of XDR—*K. pneumoniae*. We observed that the minimum concentration of NAC required for killing the exponentially growing *K. pneumoniae* ranged between 5 and 10 mg/ml, and a significant number of viable cells were recovered from the cultures treated with NAC concentrations <5 mg/ml. In 2011, Darouiche's group also reported that the MIC and MBIC of NAC against the clinical isolates of *K. pneumoniae* were 5 and 10 mg/ml, respectively (Magesh et al. 2013).

Recently, several groups reported that NAC was bactericidal at a concentration as low as 2 mg/ml (Kundukad et al. 2020). This may be attributed to the strain-specific antibacterial activity of NAC or probably clinical isolates have comparatively higher tolerance to NAC. Moreover, our cell viability results indicated that NAC is bactericidal to *K. pneumoniae* isolates, contradicting previous studies that reported NAC as bacteriostatic (Nirwati et al. 2019). However, few other studies have demonstrated that NAC is bacteriostatic at low concentrations and bactericidal at higher concentrations (Cusumano et al. 2019). We then assessed the antibiofilm activity of NAC at different concentrations against preformed 96-h-old mature biofilms of *K. pneumoniae*. Although *in vitro* biofilms of 16.6% of isolates and of 8.3% of isolates were eradicated at 10 and 20 mg/ml, respectively, complete eradication of biofilms of most of the XDR-*K. pneumoniae* isolates could be achieved alone at a concentration of 100 mg/ml.

Although the clinical use of acidic compounds may be potentially toxic, our choice of NAC as an alternative therapeutic to antibiotics is persuasive, as several reports suggested the lack of toxicity of intravenous administration of NAC to patients in various clinical situations (Aiyer et al. 2021).

In addition, NAC acts as an ROS scavenger and a cysteine prodrug that releases cysteine within cells at a slow rate, helping to restore glutathione levels (Manoharan et al. 2020). NAC showed preventive cytotoxicity effects on U937 monocytes and human kidney epithelial cells (HK-2 cell line) (Prescott et al. 1979). Similarly, we showed that treating mouse fibroblast cells (L929) and macrophage cells (J774A.1) with NAC up to a concentration of 80 mg/ml had no cytotoxicity even after 30 mins of exposure.

RNA seq data provided crucial insights on the genomic expression modulated post-NAC treatment. Among the DEG, ~80% of the genes were downregulated, indicating a catastrophic impact of NAC on bacteria. The upregulated genes were associated with env. information processing proteins; associated with the essential process for sensing and adapting

to environmental circumstances shifts. As an example, ATP-binding cassette (ABC) protein genes were found to be highly upregulated, which are involved in the efflux of xenobiotics (Greene et al. 2018). Upregulation of the ABC transporter system is a probable indication of the attempt to stabilize the homeostasis by the microorganism. Several tRNAs were found upregulated, which can be linked to early stage resistance development against antimicrobial agents (Fang et al. 2022). So, it is safe to conclude that the upregulated genes are to express proteins predominantly involved in the refurbishment and repairment of cells under chemical stress. Whereas downregulated genes are linked with different biogenesis processes, like oxidoreductase activity, menaquinone biosynthesis, histidine biosynthesis, and pyridoxine biosynthesis. The result showed that the menaquinone synthesis pathway is highly modulated, which is a crucial molecule required in the bacterial electron transport chain (Kurosu and Begari 2010). Genes of several enzymes of menaquinone synthesis pathways were affected by NAC treatment with very high significance. Pathway enrichment data delineate the modulation of metabolism pathways involving a large set of interconnected genes. However, we found that a more specific affected genes by NAC were the ribosomal proteins and translation machinery, which do have direct involvement in any other biosynthesis pathways. Further analysis showed that a cluster of genes associated with the ribosome were significantly downregulation. Functional gene enrichment analysis and KEGG pathway analysis also corroborated our observations, emphasizing that the downregulation of this particular gene cluster has impacted several aspects of cellular functions and molecular processes, resulting in reduced cellular components and, hence, viability of XDR *K. pneumoniae*. Seventeen ribosomal large subunit proteins, nine small subunit proteins, two rRNAs, ribosome-binding factor A, and a ribosome recycling factor were significantly downregulated. All of these proteins directly participate in the translation process; hence, a negative modulation of translation was seen due to NAC treatment. Further, the STRING analysis pointed toward a prominently affected chain of ribosomal proteins, supported by a high confidence score, signifying a substantial influence on genes that display co-expression and co-occurrence resulting in significant impairment of ribosomal function. Using the STRING database, initial layer of interaction analysis of most downregulated genes revealed that NAC treatment predominantly affected both the large and small subunits of ribosomal proteins. Computational analysis showed that all five analyzed genes play a crucial role in various functional networks. Specifically, their downregulation had a profound impact on processes related to the control of ribosomal subunit biogenesis, translation, binding of ribosomal protein to rRNA, and the composition of ribosomal subunits. This suggests that the suppression of these genes may have a cascading effect on the regulation and functioning of the ribosome, a fundamental cellular machinery responsible for protein synthesis. We understand that transcriptomic studies would normally need a confirmation by QRT-PCR. However, it is important to note that the transcriptomic techniques such as DNA microarray places emphasis on the reproducibility of data, necessitating validation through qPCR. In contrast, RNA-seq does not encounter the same challenges. Numerous research articles have extensively addressed the correlation between results obtained with RNA-seq and qPCR, supporting the robustness of RNA-seq methodology (Coenye 2021).

Our RNA-seq analysis was done in triplicate to make the data robust. Hence, we did not feel the need to carry out the QRT-PCR.

Previous studies have indicated that NAC reduces EPS production during biofilm formation, resulting in a noticeable change in the texture of the biofilm (Olofsson *et al.* 2003). NAC treatment on bacteria has been observed to downregulate anaerobic respiration, thereby making the microorganism less fit for adopting a biofilm lifestyle. It is well-studied that in biofilms, not all metabolic processes slow down, especially the genes crucial for supplementing the survival of the bacteria residing in biofilm, undergoing chemical and nutritional stress (Becker *et al.* 2001). Our study has uncovered that NAC treatment on *K. pneumoniae* has a detrimental impact on bacterial ribosome assembly and function, resulting in the repression of a range of gene expressions, including those pivotal for the bacterial survival during the biofilm mode, such as the bacterial secretion system and quorum sensing.

Conclusion

We summarize that physiologically tolerable concentrations of NAC can completely eradicate the *in vitro* mature biofilms of all tested XDR-*K. pneumoniae* clinical isolates grown in the TIVAP. To investigate the effect of NAC treatment on gene expression of *K. pneumoniae*, RNA-seq analysis was carried out. The transcriptome data demonstrated that NAC administration dramatically reduced a range of enzymes involved in several biogenesis processes, such as *K. pneumoniae* oxidoreductase activity, menaquinone biosynthesis, histidine biosynthesis, uridine nucleosidase, and pyridoxine biosynthesis. Our results contributed to uncover the anti-biofilm activity of the NAC against *K. pneumoniae* and we believe that NAC deserves further preclinical and clinical evaluations for effective management of WHO critical priority pathogen XDR-*K. pneumoniae*-associated nosocomial infections.

Acknowledgements

AC would like to thank UGC-BSR (F.30-487/2019(BSR)), DST-Nanomission (DST/NM/NB/2018/203), ICMR (OMI/20/2020-ECD-1) and SERB-CRG (CRG/2021/001974) for the funding support. All the experiments were approved by the Institutional Biosafety Committee, Tripura University as per guidelines of Department of Biotechnology, Govt. of India.

Supplementary data

Supplementary data is available at JAMBIO Journal online.

Conflict of interest: The authors have no conflict of interest to declare.

Author contributions

Ankurita Bhowmik (Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft), Sambuddha Chakraborty (Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft), Anusha Rohit (Project administration, Resources, Supervision, Writing – review & editing), and Ashwini Chauhan (Conceptualization, Funding acquisition,

Project administration, Resources, Supervision, Validation, Writing – review & editing)

References

- Aiyer A, Visser SK, Bye P *et al.* Effect of N-acetylcysteine in combination with antibiotics on the biofilms of three cystic fibrosis pathogens of emerging importance. *Antibiotics* 2021;10:1176. <https://doi.org/10.3390/antibiotics10101176>
- Andrews S. FastQC A Quality Control tool for High Throughput Sequence Data. 2017. <https://www.scienceopen.com/book?vid=46d7fada-29bb-41f8-9368-65453cd2c7c2> (3 May 2017, date last accessed).
- Bao J, Wu N, Zeng Y *et al.* Non-active antibiotic and bacteriophage synergism to successfully treat recurrent urinary tract infection caused by extensively drug-resistant *Klebsiella pneumoniae*. *Emerg Microbes Infect* 2020;9:771–4. <https://doi.org/10.1080/22221751.2020.1747950>
- Becker P, Hufnagle W, Peters G *et al.* Detection of differential gene expression in biofilm-forming versus planktonic populations of *Staphylococcus aureus* using micro-representational-difference analysis. *Appl Environ Microb* 2001;67:2958–65. <https://doi.org/10.1128/AEM.67.7.2958-2965.2001>
- Chauhan A, Lebeaux D, Decante B *et al.* A rat model of central venous catheter to study establishment of long-term bacterial biofilm and related acute and chronic infections. *PLoS One* 2012;7:e37281. <https://doi.org/10.1371/journal.pone.0037281>
- Coenye T. Do results obtained with RNA-sequencing require independent verification? *Biofilm* 2021;3:100043.
- Cusumano JA, Caffrey AR, Daffinee KE *et al.* Weak biofilm formation among carbapenem-resistant *Klebsiella pneumoniae*. *Diagn Microbiol Infect Dis* 2019;95:114877. <https://doi.org/10.1016/j.diagmicrobio.2019.114877>
- Daikos GL, Tsaousi S, Tzouveleki LS *et al.* Carbapenemase-producing *Klebsiella pneumoniae* bloodstream infections: lowering mortality by antibiotic combination schemes and the role of carbapenems. *Antimicrob Agents Chemother* 2014;58:2322–8. <https://doi.org/10.1128/AAC.02166-13>
- Effah CY, Sun T, Liu S *et al.* *Klebsiella pneumoniae* : an increasing threat to public health. *Ann Clin Microbiol Antimicrob* 2020;19:1.
- Ewels P, Magnusson M, Lundin S *et al.* MultiQC: summarize analysis results for multiple tools and samples in a single report. *Bioinformatics* 2016;32:3047–8. <https://doi.org/10.1093/bioinformatics/btw354>
- Fang H, Zeng G, Gu W *et al.* Genome recombination-mediated tRNA up-regulation conducts general antibiotic resistance of bacteria at early stage. *Front Microbiol* 2022;12:1–13. <https://doi.org/10.3389/fmicb.2021.793923>
- Geetha PV, Aishwarya KVL, Mariappan S *et al.* Fluoroquinolone resistance in clinical isolates of *Klebsiella pneumoniae*. *J Lab Phys* 2020;12:121–5. <https://doi.org/10.1055/s-0040-1716478>
- Greene NP, Kaplan E, Crow A *et al.* Antibiotic resistance mediated by the MacB ABC transporter family: a structural and functional perspective. *Front Microbiol* 2018;9:369512. <https://doi.org/10.3389/fmicb.2018.00950>
- Guilhen C, Miquel S, Charbonnel N *et al.* Colonization and immune modulation properties of *Klebsiella pneumoniae* biofilm-dispersed cells. *npj Biofilms Microbiomes* 2019;5:1–11. <https://doi.org/10.1038/s41522-019-0098-1>
- Hirshfield IN, Terzulli S, O'Byrne C. Weak organic acids: a panoply of effects on bacteria. *Sci Prog* 2003;86:245–69. <https://doi.org/10.3184/003685003783238626>
- Huang da W, Sherman BT, Lempicki RA. Systematic and integrative analysis of large gene lists using DAVID bioinformatics resources. *Nat Protoc* 2009;4:44–57. <https://doi.org/10.1038/nprot.2008.211>
- Indian Council of Medical Research. Antimicrobial Resistance Research and Surveillance Network January 2020 to December 2020. 2020. https://main.icmr.nic.in/sites/default/files/guidelines/AMRSN_annual_report_2020.pdf (20 April 2023, date last accessed).

- Joseph L, Mercieca T, Forestier C *et al.* From *Klebsiella pneumoniae* colonization to dissemination: an overview of studies implementing murine models. *Microorganisms* 2021;9:1282. <https://doi.org/10.3390/microorganisms9061282>
- Jun Y, Youn CK, Jo ER *et al.* In vitro inhibitory activity of N-acetylcysteine on tympanostomy tube biofilms from methicillin-resistant *Staphylococcus aureus* and quinolone-resistant *Pseudomonas aeruginosa*. *Int J Pediatr Otorhinolaryngol* 2019;126:109622. <https://doi.org/10.1016/j.ijporl.2019.109622>
- Kundukad B, Udayakumar G, Grela E *et al.* Weak acids as an alternative anti-microbial therapy. *Biofilm* 2020;2:100019. <https://doi.org/10.1016/j.biofilm.2020.100019>
- Kurosu M, Begari E. Vitamin K2 in electron transport system: are enzymes involved in vitamin K2 biosynthesis promising drug targets? *Molecules* 2010;15:1531–53. <https://doi.org/10.3390/molecules15031531>
- Langmead B, Salzberg SL. Fast gapped-read alignment with Bowtie 2. *Nat Methods* 2012;9:357–9. <https://doi.org/10.1038/nmeth.1923>
- Lebeaux D, Chauhan A, Létoffé S *et al.* pH-mediated potentiation of aminoglycosides kills bacterial persisters and eradicates *in vivo* biofilms. *J Infect Dis* 2014;210:1357–66.
- Lim G, Lim CJ, Lee JH *et al.* Identification of new target proteins of a Urotensin- II receptor antagonist using transcriptome-based drug repositioning approach. *Sci Rep* 2021;11:1–11. <https://doi.org/10.1038/s41598-021-96612-0>
- Magesh H, Kumar A, Alam A *et al.* Identification of natural compounds which inhibit biofilm formation in clinical isolates of *Klebsiella pneumoniae*. *Indian J Exp Biol* 2013;51:764–72. <https://pubmed.ncbi.nlm.nih.gov/24377137/>
- Manoharan A, Das T, Whiteley GS *et al.* The effect of N-acetylcysteine in a combined antibiofilm treatment against antibiotic-resistant *Staphylococcus aureus*. *J Antimicrob Chemother* 2020;75:1787–98. <https://doi.org/10.1093/jac/dkaa093>
- NIH U.S. National Library of Medicine. ACETYLCYSTEINE-acetylcysteine injection, solution Fresenius Kabi USA, LLC. 2018. <https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=8cca4e4c-85a2-d575-e053-2995a90a6599> (12 June 2022, date last accessed).
- Nirwati H, Sinanjung K, Fahrurissa F *et al.* Biofilm formation and antibiotic resistance of *Klebsiella pneumoniae* isolated from clinical samples in a tertiary care hospital, Klaten, Indonesia. *BMC Proc* 2019;13:20. <https://doi.org/10.1186/s12919-019-0176-7>
- Olofsson AC, Hermansson M, Elwing H. N-acetyl-L-cysteine affects growth, extracellular polysaccharide production, and bacterial biofilm formation on solid surfaces. *Appl Environ Microb* 2003;69:4814–22. <https://doi.org/10.1128/AEM.69.8.4814-4822.2003>
- Prescott LF, Illingworth RN, Critchley JA *et al.* Intravenous N-acetylcysteine: the treatment of choice for paracetamol poisoning. *BMJ* 1979;2:1097–100. <https://doi.org/10.1136/bmj.2.6198.1097>
- Shein AMS, Wannigama DL, Higgins PG *et al.* Novel colistin-EDTA combination for successful eradication of colistin-resistant *Klebsiella pneumoniae* catheter-related biofilm infections. *Sci Rep* 2021;11:21676. <https://doi.org/10.1038/s41598-021-01052-5>
- Tenorio MCDS, Graciliano NG, Moura FA *et al.* N-acetylcysteine (NAC): impacts on human health. *Antioxidants* 2021;10:967.
- van der Weide H, Vermeulen-de Jongh D, van der Meijden A *et al.* Antimicrobial activity of two novel antimicrobial peptides AA139 and SET-M33 against clinically and genotypically diverse *Klebsiella pneumoniae* isolates with differing antibiotic resistance profiles. *Int J Antimicrob Agents* 2019;54:159–66. <https://doi.org/10.1016/j.ijantimicag.2019.05.019>
- Vecchiarelli A, Dottorini M, Pietrella D *et al.* Macrophage activation by N-acetyl-cysteine in COPD patients. *Chest* 1994;105:806–11. <https://doi.org/10.1378/chest.105.3.806>
- WHO. WHO publishes list of bacteria for which new antibiotics are urgently needed. 2017. <https://www.who.int/news/item/27-02-2017-who-publishes-list-of-bacteria-for-which-new-antibiotics-are-urgently-needed> (19 April 2023, date last accessed)